Dang Minh Quan

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github.com/dangminhquandk | Computer Science Student (IT1) – HUST

Professional Summary

High-achieving Computer Science student at Hanoi University of Science and Technology (HUST) maintaining an outstanding **CPA of 3.77 / 4.00 (Top-tier academic standing)**. Possesses a rigorous foundation in Applied Algorithms, Low-Level System Mechanics, Database Architecture, and hands-on research in **Distributed Computing (Apache Spark & GraphFrames)**. Demonstrates exceptional capacity for deep analytical problem-solving and memory-conscious engineering. Highly driven to join TPBank as a **Fresh Data Engineer**, leveraging distributed computing and algorithmic rigor to architect high-throughput, fault-tolerant financial data pipelines.

Education

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam
Candidate for Bachelor of Science in Computer Science (IT1), CPA: 3.77 / 4.00 *Sep 2024 – 2028*

- **Academic Honors:** Cumulative Pass Average (CPA): **3.77 / 4.00** — Solid theoretical foundation across core computer science and software engineering curricula.
- **Selected Coursework:** Data Structures & Algorithms (A), Object-Oriented Programming (A+), Computer Networks (A+), Operating Systems (A), Applied Algorithms (A), Introduction to AI (A), Probability & Statistics (A+), Calculus I–III (A), Technical Writing & Presentation (A+).

Technical Skills

- **Distributed Systems & Big Data:** Apache Spark, PySpark, GraphFrames, Graph Analytics, ETL Pipeline Design, RDD/DataFrame Optimization.
- **Programming Languages:** Python, C/C++, Java, SQL (PostgreSQL, MySQL).
- **Core CS & Systems Architecture:** Data Structures & Algorithms, Low-Level Memory Management, System Architecture, Object-Oriented Programming (OOP), Data Preprocessing & Imbalance Handling.
- **Frameworks & Developer Tools:** NumPy, Pandas, Scikit-Learn, Streamlit, Joblib, Docker, Git, GitHub, Linux/Bash, VS Code, LaTeX.

Projects & Research Experience

Graph Analytics & Distributed Computing Research *2026 – Present*
Undergraduate Research Assistant | SEDS Lab, HUST *PySpark, GraphFrames, Spark SQL*

- Collaborating on research with lab faculty regarding distributed graph processing frameworks (**PySpark** & **GraphFrames**) for large-scale transaction networks.
- Implemented hands-on prototypes to explore motif finding (`g.find()`) for anomaly detection, as well as core algorithms (`Connected Components` for Entity Resolution and `PageRank`).
- Researched Spark execution mechanics, memory management, and DAG lineage truncation (`setCheckpointDir`) to optimize iterative graph computations.

Production-Ready Text Classification & Preprocessing Pipeline *2025*
Core Developer & Presenter | Course: Intro to AI *Python, Streamlit, NumPy, Joblib*
github.com/dangminhquandk/Email_Spam_Classification

- Programmed core Naive Bayes (Laplace smoothing and log-likelihood math) and KNN completely from scratch via NumPy to master mathematical foundations and low-level mechanics without black-box abstractions.
- Built an automated data cleaning and ingestion workflow; resolved severe class imbalance through Random Under-Sampling and deduplication to establish a balanced 50:50 training dataset.
- Benchmarked system design trade-offs: achieved 95.14% accuracy and 0.95 F1-score with Naive Bayes; deliberately excluded the memory-heavy KNN model (>110MB footprint) from production deployment to optimize server resource efficiency.
- Served real-time inference through an interactive, decoupled Streamlit web graphical user interface.

High-Performance Algorithmic Repository & System Design *2026*
Independent Developer *C++, System Design, Git*
github.com/dangminhquandk/Data-Structures-Algorithms-CPA374

- Architected a scalable C++ codebase categorizing advanced algorithmic paradigms: Graph Theory, Dynamic Programming, and Custom Data Structures with strict time/space complexity bounds ($\mathcal{O}(V+E)$, $\mathcal{O}(N \log N)$).
- Enforced robust pointer mechanics and low-level memory allocation protocols to simulate hardware-constrained environments, ensuring optimal cache locality and execution speed.

Languages

- **Vietnamese:** Native Proficiency.
- **English:** Professional Technical Reading (Capacity to parse distributed system whitepapers, official open-source documentation, and RFC specifications).